Replicating a Experimental Study (THE I - Advanced)

Research Design and Methods in Quantitative Research

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Instructions

Please read and follow the guidelines below carefully. Then, complete the exercises and report the results in a Quarto document. Compile the Quarto document in PDF and submit both the compiled PDF and `.qmd` files within the deadline.

Further instructions about the submission are below.

Preparation step 1: Install R and RStudio

To complete this exercise, you will need **R** and **RStudio**. If you don't have them installed already, download and install them from:

- R
- RStudio

A tutorial on how to start using R and R Studio is here. Please **contact the lecturer** and collaborate with your classmates if you have any doubts or need help.

Preparation step 2: Prepare a Quarto Document

Open RStudio, create a new Quarto document (`.qmd`), and set the output format to PDF. Make sure your Quarto installation is up-to-date:

```
# Install Quarto if needed
## Run this line in a separate script or the Quarto document will not compile
# install.packages("quarto")
```

You can find help on how to set up a Quarto document here.

Preparation step 3: Read the Assigned Paper and Download the Replication Files

You will need to read the paper *Instrumentally Inclusive: The Political Psychology of Homonationalism*, which you can find in the exercise materials folder.

After reading the paper, download the Study 1 replication materials, also provided in the materials folder:

- Data file: `study1_data.csv`
- R script: `study1.R`

These files have been downloaded from the authors' replication repository. Take a moment to explore it—next time, you may need to download the files yourself!

If you look at their repository, you will notice that the replication materials **do not include a codebook** for the data, so you will need to navigate their R script to identify the relevant variables.

Exercises

Exercise 1: Summary of the Paper and Main Findings

Provide a brief summary of the paper and describe its main findings, especially those related to Study 1. Remember to be concise—two or three paragraphs are usually enough for a good summary.

Exercise 2: Data Exploration

Load the data file `study1_data.csv` into R and describe the dataset. Then, describe the experimental variables `treatment` and `imm_1` (i.e., pre-treatment attitudes toward immigration) by providing visualizations of their distributions. Next, select up to four covariates (e.g., gender, age, etc.) and plot their distributions as well. What do these visualizations tell us about the data?

Importantly, you may need to clean (e.g., recode or remove missing values) or transform (e.g., convert a numeric into a categorical variable) some of the variables to produce appropriate visualizations. Document any changes made.

Exercise 3: Check Covariate Balance

Check for balance across covariates by examining whether the mean values and distributions of your selected covariates are similar between the treatment and control groups. Report the results in a table *and* with plots.

Explain your findings. Why would you expect randomization to lead to balance across covariates?

Exercise 4: Replicate the Main Finding by Varying Sample Size

Estimate the average effect of the treatment (ATE) on the outcome variable `support` conditional on the pre-treatment immigration attitudes `imm_1`. To do this, run the following script (change variable names if you made any modifications earlier):

```
# Run an OLS model with int. between treatment and immigration attitudes
## Change 'df' with the name you gave to the dataset
# model_study1 <- lm(support ~ treatment * imm_1, data = df)

# Check the results
# summary(model_study1)
```

The code line you just ran estimates an OLS regression with an interaction term between the experimental treatment and pre-treatment attitudes toward immigration, exactly as reported in the original paper. Compare these results with those of the authors.

Next, repeat the analysis using three progressively smaller random samples of the treatment and control groups (n = 200, n = 50, and n = 5 per group), resulting in total sample sizes of N = 400, 100, and 10, respectively.

Explain your findings: does the sample size affect the results? Why?

Submission guidelines

Please submit both the PDF file and the `.qmd` file. Both files should report all the code used for analysis and annotations explaining each step.

The name of the files must follow the structure *take-home_exercise_i_advanced_YOURSURNAME(S).pdf* and *take-home_exercise_i_advanced_YOURSURNAME(S).qmd*, respectively. They should be uploaded to the folder *Students responses/Take-home exercises/Take-home exercise I* in OLAT.

Deadline: **29.10.26**

References

Turnbull-Dugarte, S. J., & Ortega, A. L. (2024). Instrumentally inclusive: the political psychology of homonationalism. *American Political Science Review, 118(3)*, 1360-1378.